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(54) **HOT ROLLED AND STEEL SHEET AND A
METHOD OF MANUFACTURING THEREOF**

(71) Applicant: **ArcelorMittal**, Luxembourg (LU)

(72) Inventors: **Narayan POTTORE**, Munster, IN
(US); **Ranbir JAMWAL**, Chicago, IL
(US); **Brian LIN**, Hinsdale, IL (US);
Evgueni I. POLIAK, Munster, IN (US)

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ABSTRACT

A hot rolled steel sheet having a composition including of the following elements $0.060\% \leq \text{Carbon} \leq 0.1\%$, $1.4\% \leq \text{Manganese} \leq 1.8\%$, $0.3\% \leq \text{Silicon} \leq 0.8\%$, $0.01\% \leq \text{Aluminum} \leq 0.1\%$, $0.01\% \leq \text{Niobium} \leq 0.09\%$, $0.01\% \leq \text{Titanium} \leq 0.09\%$, $0.002\% \leq \text{Phosphorus} \leq 0.02\%$, $0\% \leq \text{Sulfur} \leq 0.005\%$, $0\% \leq \text{Nitrogen} \leq 0.01\%$, with $0.045\% \leq \text{Ti+Nb} \leq 0.060\%$, $0\% \leq \text{Chromium} \leq 0.5\%$, $0\% \leq \text{Molybdenum} \leq 0.4\%$, $0\% \leq \text{Vanadium} \leq 0.4\%$, $0\% \leq \text{Calcium} \leq 0.005\%$, $0\% \leq \text{Copper} \leq 1\%$, $0\% \leq \text{Nickel} \leq 1\%$, $0\% \leq \text{Boron} \leq 0.05\%$, $0\% \leq \text{Magnesium} \leq 0.05\%$, the remainder composition being composed of iron and unavoidable impurities caused by processing, the microstructure of the steel sheet including in area fraction, 10% to 20% Bainite, 80% to 92% Ferrite wherein the aspect ratio of the grains of Bainite and Ferrite is less than 1.75 and a cumulated amount of Residual Austenite and Martensite is from 0% to 10%.

HOT ROLLED AND STEEL SHEET AND A METHOD OF MANUFACTURING THEREOF

[0001] The present invention relates to hot rolled steel sheets suitable for use as steel sheet for automobiles.

BACKGROUND

[0002] Automotive parts are required to satisfy two inconsistent necessities, e.g. ease of forming and strength, but in recent years a third requirement of improvement in fuel consumption is also bestowed upon automobiles in view of global environment concerns. Thus, now automotive parts must be made of material having high formability in order that to fit in the criteria of ease of fit in the intricate automobile assembly and at same time have to improve strength for vehicle crashworthiness and durability while reducing weight of vehicle to improve fuel efficiency.

SUMMARY OF THE INVENTION

[0003] Therefore, intense Research and development endeavors are put in to reduce the amount of material utilized in car by increasing the strength of material. Conversely, an increase in strength of steel sheets decreases formability, and thus development of materials having both high strength and high formability is necessitated.

[0004] Earlier research and developments in the field of high strength and high formability steel sheets have resulted in several methods for producing high strength and high formability steel sheets, some of which are enumerated herein for conclusive appreciation of the present invention:

[0005] CN113201691 discloses a 590 MPa-level hot rolled steel plate for hydraulic bulging and a preparation method thereof, and belongs to the technical field of steel material engineering. The 590 MPa-grade hot rolled steel plate for hydraulic bulging comprises the following chemical components in percentage by weight: 0.02-0.04% of C, less than or equal to 0.05% of Si, 1.05-1.15% of Mn, 0.040-0.050% of Nb, 0.050% of Ti, 0.30-0.60% of Cr, 0.02% of P, less than or equal to 0.02% of S, less than or equal to 0.003% of N, 0.010-0.050% of Al, 0.0010-0.0050% of Ca, and the balance of Fe and inevitable impurities. The invention adopts a micro/low C—Nb—Ti—Cr microalloying component system and obtains the microstructure of ultrafine grained acicular ferrite and a nano precipitated phase by a controlled rolling and controlled cooling process technology. The mechanical properties of the steel plate provided by the invention meet the following requirements: the yield strength is more than or equal to 500 MPa, the tensile strength is more than or equal to 590 MPa, the elongation is more than or equal to 25 percent, the n value is more than or equal to 0.12, the d-a is qualified in a 180-degree cold bending test, and the grain size is more than or equal to 12 grade, so that the problem of poor forming performance of the existing high-strength steel can be effectively solved. However CN113201691 does not exhibit homogenous properties in all three directions that are transversal direction, longitudinal direction and diagonal direction therefore limits its usage in the automotive industry.

[0006] It is an object of the present invention to solve these problems by making available hot-rolled steel sheets that simultaneously have:

[0007] an ultimate tensile strength greater than or equal to 590 MPa in three directions that are transversal

direction, longitudinal direction and diagonal direction and preferably greater than or equal to 600 MPa in three directions that are transversal direction, longitudinal direction and diagonal direction,

[0008] a yield strength from 500 MPa to 600 MPa in three directions that are transversal direction, longitudinal direction and diagonal direction and preferably a yield strength from 510 MPa to 580 MPa in three directions that are transversal direction, longitudinal direction and diagonal direction,

[0009] a total elongation greater than or equal to 20% in three directions that are transversal direction, longitudinal direction and diagonal direction and preferably a total elongation greater than or equal to 22% in three directions that are transversal direction, longitudinal direction and diagonal direction, and

[0010] a hole expansion ratio of greater than or equal to 75% and preferably greater than or equal to 77%.

[0011] The present invention provides a hot rolled steel sheet having a composition comprising of the following elements, expressed in percentage by weight:

$0.060\% \leq \text{Carbon} \leq 0.1\%$;

$1.4\% \leq \text{Manganese} \leq 1.8\%$;

$0.3\% \leq \text{Silicon} \leq 0.8\%$;

$0.01\% \leq \text{Aluminum} \leq 0.1\%$;

$0.01\% \leq \text{Niobium} \leq 0.09\%$;

$0.01\% \leq \text{Titanium} \leq 0.09\%$;

$0.002\% \leq \text{Phosphorus} \leq 0.02\%$;

$0\% \leq \text{Sulfur} \leq 0.005\%$;

$0\% \leq \text{Nitrogen} \leq 0.01\%$;

with

$0.045\% \leq \text{Ti} + \text{Nb} \leq 0.060\%$;

and one or more of the following optional elements:

$0\% \leq \text{Chromium} \leq 0.5\%$;

$0\% \leq \text{Molybdenum} \leq 0.4\%$;

$0\% \leq \text{Vanadium} \leq 0.4\%$;

$0\% \leq \text{Calcium} \leq 0.005\%$;

$0\% \leq \text{Copper} \leq 1\%$;

$0\% \leq \text{Nickel} \leq 1\%$;

$0\% \leq \text{Boron} \leq 0.05\%$;

$0\% \leq \text{Magnesium} \leq 0.05\%$;

the remainder composition being composed of iron and unavoidable impurities caused by processing, the microstructure of steel sheet comprising, by area percentage, 10% to 20% Bainite, 80% to 92% Ferrite wherein the aspect ratio of the grains of Bainite and Ferrite is less than 1.75 and a cumulated amount of Residual Austenite and Martensite is from 0% to 10%.

[0012] In a preferred embodiment, the steel sheets according to the invention may also present a yield strength to tensile strength ratio of 0.5 or more.

[0013] Preferably, such steel can also have a good suitability for forming, in particular for rolling with good weldability and coatability.

[0014] Another object of the present invention is to make available a method for the manufacturing of these sheets that is compatible with conventional industrial applications while being robust towards manufacturing parameters shifts.

DETAILED DESCRIPTION

[0015] The hot rolled steel sheet of the present invention may optionally be coated with zinc or zinc alloys, to improve its corrosion resistance.

[0016] Carbon is present in the steel from 0.060% to 0.1%. Carbon is an element necessary for imparting the mechanical properties to the steel of the present invention by facilitating the formation of ferrite and carbon also impart the steel with strength by precipitate strengthening by forming Titanium Carbide or Niobium Carbides, therefore, Carbon plays a pivotal role in increasing the strength. But Carbon content less than 0.06% will not be able to impart the tensile strength to the steel of the present invention. On the other hand, at a Carbon content exceeding 0.1%, the steel exhibits poor spot weldability which limits its application for the automotive parts. A preferable content for the present invention may be kept from 0.065% to 0.1% and more preferably from 0.07% to 0.095%.

[0017] Manganese content of the steel of the present invention is from 1.4% to 1.8%. This element is gamma-gamma and influences Bs and Ms temperatures therefore plays an important role in controlling the bainite formation. The purpose of adding Manganese is essentially to impart hardenability to the steel. An amount of at least 1.4% by weight of Manganese has been found in order to provide the strength and hardenability to the steel sheet. But when Manganese content is more than 1.8% it produces adverse effects such as it retards transformation of Austenite during the cooling after hot rolling. In addition, the Manganese content of above 1.8% it promotes the central segregation hence reduces the formability and deteriorates the weldability of the present steel. A preferable content for the present invention may be kept from 1.45% to 1.7% and more preferably from 1.45% to 1.65%.

[0018] Silicon content of the steel of the present invention is from 0.3% to 0.8%. Silicon is a solid solution strengthener especially for microstructures Ferrite and Bainite. In addition, a higher content of Silicon can retard the precipitation of Cementite. However, disproportionate content of Silicon leads to a problem such as surface defects like tiger strips which adversely effects the coatability of the steel of the present invention. Therefore, the concentration is controlled within an upper limit of 0.8%. A preferable content for the present invention may be kept from 0.35% to 0.75% and more preferably from 0.35% to 0.6%.

[0019] Aluminum is an element that is present in the steel of the present invention from 0.01% to 0.1%. Aluminum is an alpha-gamma element and imparts ductility to steel of the present invention. Aluminum in the steel has a tendency to bond with nitrogen to form aluminum nitride hence from point of view of the present invention the Aluminum content must be kept as low as possible and preferably from 0.02% to 0.06%.

[0020] Niobium is an essential element for the present invention. Niobium content may be present in the steel of the present invention from 0.01% to 0.09% and is added in the steel of the present invention for forming carbides or carbo-nitrides to impart strength to the steel of the present invention by precipitation strengthening. A preferable content for the present invention may be kept from 0.02% to 0.07% and more preferably from 0.02% to 0.05%.

[0021] Titanium is also an essential element which added to the steel of the present invention from 0.01% to 0.09%, preferably from 0.01% to 0.06%. As niobium, it is involved in the formation of carbides and carbo-nitrides so it plays a role in hardening and providing strength to the steel. But it is also involved to form TiN appearing during the solidification of the cast product. The amount of Ti is limited to 0.09% to avoid coarse TiN detrimental for the hole expansion. In case the titanium content is below 0.01% it does not impart any effect on the steel of the present invention.

[0022] Phosphorus constituent of the steel of the present invention is from 0.002% to 0.02%. Phosphorus reduces the spot weldability and the hot ductility, particularly due to its tendency to segregate at the grain boundaries or co-segregate with manganese. For these reasons, its content is limited to 0.02% and preferably lower than 0.015%.

[0023] Sulfur is not an essential element but may be contained as an impurity in steel and the from point of view of the present invention the Sulfur content is preferably as low as possible, but is 0.005% or less from the viewpoint of manufacturing cost. Further if higher Sulfur is present in steel it combines to form Sulfides especially with Manganese and reduces its beneficial impact on the steel of the present invention, therefore preferred below 0.003%.

[0024] Nitrogen is limited to 0.01% in order to avoid ageing of material, nitrogen forms the nitrides which impart strength to the steel of the present invention by precipitation strengthening with Vanadium and Niobium but whenever the presence of nitrogen is more than 0.01% it can form high amount of Aluminum Nitrides which are detrimental for the present invention hence the preferable upper limit for nitrogen is 0.005%.

[0025] Chromium is an optional element for the present invention. Chromium content may be present in the steel of the present invention is from 0% to 0.5%. Chromium is an element that provides hardenability to the steel but higher content of Chromium higher than 0.5% leads to central co-segregation similar to Manganese.

[0026] Molybdenum is an optional element that constitutes 0% to 0.4% of the Steel of the present invention. Molybdenum increases the hardenability of the steel of the present invention and influences the transformation of austenite to Ferrite and Bainite during cooling after hot rolling. However, the addition of Molybdenum excessively increases the cost of the addition of alloy elements, so that for economic reasons its content is limited to 0.4%. Preferable limit for molybdenum is from 0% to 0.3%.

[0027] Vanadium is an optional element that constitutes from 0% to 0.4% of the steel of the present invention. Vanadium is effective in enhancing the strength of the steel by forming carbides, nitrides or carbo-nitrides and the upper limit is 0.4% due to the economic reasons. These carbides, nitrides or carbo-nitrides are formed during the second and third step of cooling. Preferable limit for Vanadium is from 0% to 0.3%.

[0028] Nickel may be added as an optional element in an amount of 0% to 1% to increase the strength of the steel and to improve its toughness. A minimum of 0.01% is required to produce such effects. However, when its content is above 1%, Nickel causes ductility deterioration.

[0029] Copper may be added as an optional element in an amount of 0% to 1% to increase the strength of the steel and to improve its corrosion resistance. A minimum of 0.01% is required to produce such effects. However, when its content is above 1%, copper causes hot ductility deterioration during hot rolling.

[0030] Calcium is an optional element which may be added to the steel of the present invention up to 0.005%, preferably from 0.001% to 0.005%. Calcium is added to the steel of the present invention as an optional element especially during the inclusion treatment. Calcium contributes towards the refining of the steel by arresting the detrimental sulphur content in globularizing it.

[0031] Other elements such as cerium, boron, magnesium or zirconium can be added individually or in combination in the following proportions: $Ce \leq 0.1\%$, $B \leq 0.05\%$, $Mg \leq 0.05\%$ and $Zr \leq 0.05\%$. Up to the maximum content levels indicated, these elements make it possible to refine the grain during solidification.

$$0.045 \leq Ti + Nb \leq 0.060$$

[0032] The cumulative presence of Titanium and Niobium is kept from 0.045% to 0.060% to impart the steel of the present invention with strength and hole expansion ratio as both Niobium and Titanium form carbonitrides or carbides, hence this equation supports the present invention to strike a balance between tensile strength by ensuring formation of precipitates and imparts hole expansion ratio by ensuring adequate ferrite formation during the cooling after the hot rolling.

[0033] The remainder of the composition of the Steel consists of iron and inevitable impurities resulting from processing.

[0034] The microstructure of the Steel sheet comprises the following,

[0035] Ferrite constitutes from 80% to 92% of microstructure by area fraction for the Steel of the present invention. Ferrite constitutes the primary phase of the steel as a matrix phase. Ferrite cumulatively comprises of Polygonal ferrite and acicular ferrite. Ferrite imparts elongation as well as formability to the steel of the present invention in all three directions. To ensure an elongation of 20% or more it is necessary to have at least 80% of Ferrite. Ferrite is formed during the cooling after hot rolling in steel of the present invention. But whenever ferrite content is present above 92% in the steel of the present invention the tensile strength is not achieved. Preferably, the content of Ferrite is from 82% to 90% and more preferably from 85% to 90%.

[0036] Bainite constitutes from 10% to 20% of microstructure by area fraction for the Steel of the present invention. Bainite cumulatively comprises of Upper Bainite and Lower Bainite, Granular bainite lath bainite and Carbide free bainite. To ensure tensile strength of 590 MPa or more in all three direction it is necessary to have 10% of Bainite. Bainite starts forming from cooling stop temperature and

forms till the coiling. Preferably, the content of Bainite is from 10% to 18% and more preferably from 10% to 15%.

[0037] The cumulated amounts of bainite and ferrite is always greater than 90% to ensure a balance from strength and formability. In a preferred embodiment, cumulated amounts of bainite and ferrite is always greater than 95% and more preferably 97%. Cumulative presence of Bainite and Ferrite impart tensile strength of 590 MPa due to the presence of Bainite and Ferrite ensure the formability. Further, bainite and ferrite grains of the present invention have an aspect ratio of less than 1.75 and preferably less than 1.70 microns which in return imparts the steel with strength in all three direction while simultaneously having a hole expansion ratio of 75% or more.

[0038] Martensite and Residual Austenite are optional constituents for the steel of the present invention and may be present from 0% to 10% cumulatively by area fraction and are found in traces. Martensite for the present invention includes both fresh martensite and tempered martensite. Martensite imparts strength to the Steel of the present invention. When Martensite is in excess of 10% it imparts excess strength and the yield strength goes beyond acceptable upper limit. In a preferred embodiment, the cumulated amount of martensite and residual austenite is from 0 to 5%.

[0039] In addition to the above-mentioned microstructure, the microstructure of the hot rolled steel sheet is free from microstructural components, such as Pearlite and Cementite but may be found in traces.

[0040] A steel sheet according to the invention can be produced by any suitable method. A preferred method consists in providing a semi-finished casting of steel with a chemical composition according to the invention. The casting can be done either into ingots or continuously in form of thin slabs or thin strips, i.e. with a thickness ranging from approximately 220 mm for slabs up to several tens of millimeters for thin strip.

[0041] For example, a slab having the above-described chemical composition is manufactured by continuous casting wherein the slab optionally underwent the direct soft reduction during the continuous casting process to avoid central segregation and to ensure a ratio of local Carbon to nominal Carbon kept below 1.10. The slab provided by continuous casting process can be used directly at a high temperature after the continuous casting or may be first cooled to room temperature and then reheated for hot rolling.

[0042] The temperature of the slab, which is subjected to hot rolling, is preferably at least 1200° C. and must be below 1300° C. In case the temperature of the slab is lower than 1200° C., excessive load is imposed on a rolling mill. Therefore, the temperature of the slab is preferably sufficiently high so that hot rolling can be completed in the in 100% austenitic range. Reheating at temperatures above 1275° C. must be avoided because it causes productivity loss and is also industrially expensive. Therefore, the preferred reheating temperature is from 1200° C. to 1275° C.

[0043] Hot rolling finishing temperature for the present invention is from 850° C. to 975° C. and preferably from 880° C. to 930° C.

[0044] The hot rolled strip obtained in this manner is then cooled wherein the cooling starts immediately after the finishing of hot rolling and the hot rolled strip is cooled from finishing of hot rolling to a Cooling stop temperature range which is from 480° C. to 550° C. at a cooling rate greater

than 20° C./s and preferably the cooling rate from 40° C./s to 150° C./s and more preferably from 40° C./s to 120° C./s. The preferable cooling stop range which is from 490° C. to 540° C. and more preferably from 500° C. to 540° C.

[0045] Thereafter coiling the hot rolled strip from the coiling temperature range 480° C. and 550° C. and preferably from 490° C. to 540° C. Then cooling the coiled hot rolled strip to room temperature to obtain a hot rolled steel sheet.

EXAMPLES

[0046] The following tests, examples, figurative exemplification and tables which are presented herein are non-restricting in nature and must be considered for purposes of illustration only, and will display the advantageous features of the present invention.

[0047] Steel sheets made of steels with different compositions are gathered in Table 1, where the steel sheets are produced according to process parameters as stipulated in Table 2, respectively. Thereafter Table 3 gathers the microstructures of the steel sheets obtained during the trials and table 4 gathers the result of evaluations of obtained properties.

TABLE 1

Trails	C	Mn	Si	Al	Nb	Ti	S	P	N
I1	0.079	1.516	0.405	0.039	0.0339	0.0157	0.025	0.009	0.0018
I2	0.080	1.519	0.436	0.036	0.0356	0.0169	0.022	0.010	0.0022
I3	0.083	1.509	0.429	0.039	0.0336	0.0149	0.002	0.008	0.0037
R1	0.058	1.489	0.175	0.028	0.0633	0.0112	0.002	0.010	0.0060
R2	<u>0.054</u>	1.560	<u>0.205</u>	0.039	0.0626	0.0280	0.0046	0.013	0.0052
R3	0.080	1.525	<u>0.090</u>	0.003	0.0035	0	0.022	0.010	0.0018
R4	0.063	1.47	<u>0.280</u>	0.040	0.0367	<u>0.025</u>	0.002	0.010	0.0037

Trails	Cr	Mo	Cu	Ni	V	B	Ca	Ti + Nb
I1	0.0118	0.0005	0.0048	0.0024	0.0036	0.0002	0.0001	0.0496
I2	0.0103	0.0005	0.0043	0.0027	0.0035	0.0003	0.0002	0.0525
I3	0.0135	0.0005	0.0283	0.0130	0.0032	0.0004	0.0004	0.0485
R1	0.0153	0.0028	0.0268	0.0136	0.0021	0.0001	0	0.0745
R2	0.0190	0.0020	0.0140	0.0060	0.0020	0.0002	0.0023	<u>0.0906</u>
R3	0	0	0	0	0	0	0	<u>0.0035</u>
R4	0.04	0.004	0	0	0	0	0	<u>0.0617</u>

I = according to the invention; R = reference; underlined values: not according to the invention.

TABLE 2

Table 2 gathers the process parameters implemented on steels of Table 1.						
Trails	Reheating T (° C.)	HR Finish T (° C.)	Cooling start T (° C.)	Cooling rate (° C./s)	Cooling stop T (° C.)	Coiling T (° C.)
I1	1233	898	898	60	524	524
I2	1235	895	895	60	520	520
I3	1246	894	894	60	517	517
R1	1250	893	893	60	558	558
R2	1250	902	902	60	560	560
R3	1250	885	885	60	460	460
R4	1200	890	890	60	<u>525</u>	<u>525</u>

I = according to the invention; R = reference; underlined values: not according to the invention.

Table 3

[0048] Table 3 exemplifies the results of the tests conducted in accordance with the standards on different micro-

scopes such as Scanning Electron Microscope for determining the microstructures of both the inventive and reference steels. Aspect ratio is the ratio of the longest intercept grain dimension maximum Feret diameter(Fmax) to the longest intercept grain dimension measured at 90° of said Fmax (Fmax90°).

$$\text{Aspect Ratio} = (F_{\text{max}})/(F_{\text{max}90^\circ}).$$

[0049] The results are stipulated herein:

TABLE 4

Trails	Ferrite (%)	Bainite (%)	RA + Martensite (%)	Ferrite + Bainite(%)	Aspect Ratio of ferrite and bainite grains
I1	88	12	0	100	1.51
I2	86	13	1	99	1.52

TABLE 4-continued

Trails	Ferrite (%)	Bainite (%)	RA + Martensite (%)	Ferrite + Bainite(%)	Aspect Ratio of ferrite and bainite grains
I3	89	11	0	100	1.67
R1	<u>95</u>	<u>5</u>	0	100	<u>1.85</u>
R2	<u>95</u>	<u>5</u>	0	100	<u>1.79</u>
R3	87	12	1	99	<u>2.01</u>
R4	87	13	0	100	<u>1.77</u>

I = according to the invention; R = reference; underlined values: not according to the invention.

[0050] Table 4 exemplifies the mechanical properties of both the inventive steel and reference steels. In order to determine the tensile strength, yield strength and total elongation, tensile tests are conducted in accordance of JIS Z2241 standards.

[0051] The results of the various mechanical tests conducted in accordance to the standards are gathered

TABLE 4

Sample Steels	LONGITUDINAL DIRECTION			TRANSVERSAL DIRECTION			DIAGONAL DIRECTION			HER (%)
	UTS (MPa)	YS (MPa)	TE (%)	UTS (MPa)	YS (MPa)	TE (%)	UTS (MPa)	YS (MPa)	TE (%)	
I1	634	513	25	645	564	25	618	544	29	80
I2	644	516	24	662	561	22	628	530	25	77
I3	628	525	26	646	567	24	610	532	31	87
R1	596	500	24	613	550	23	579	520	27	81
R2	650	558	21	668	609	19	<u>622</u>	568	24	74
R3	595	475	24	611	<u>523</u>	<u>22</u>	<u>582</u>	495	28	<u>88</u>
R4	605	<u>502</u>	26	625	552	26	<u>583</u>	<u>500</u>	29	81

I = according to the invention; R = reference; underlined values: not according to the invention.

What is claimed is:

1-15. (canceled)

16. A hot rolled steel sheet having a composition comprising of the following elements, expressed in percentage by weight:

0.060% ≤ Carbon ≤ 0.1%;
 1.4% ≤ Manganese ≤ 1.8%;
 0.3% ≤ Silicon ≤ 0.8%;
 0.01% ≤ Aluminum ≤ 0.1%;
 0.01% ≤ Niobium ≤ 0.09%;
 0.01% ≤ Titanium ≤ 0.09%;
 0.002% ≤ Phosphorus ≤ 0.02%;
 0% ≤ Sulfur ≤ 0.005%;
 0% ≤ Nitrogen ≤ 0.01%;
 with
 0.045% ≤ Ti + Nb ≤ 0.060%;

and one or more of the following optional elements:

0% ≤ Chromium ≤ 0.5%;
 0% ≤ Molybdenum ≤ 0.4%;
 0% ≤ Vanadium ≤ 0.4%;
 0% ≤ Calcium ≤ 0.005%;
 0% ≤ Copper ≤ 1%;
 0% ≤ Nickel ≤ 1%;
 0% ≤ Boron ≤ 0.05%;
 0% ≤ Magnesium ≤ 0.05%;

a remainder of the composition being composed of iron and unavoidable impurities caused by processing;
 a microstructure of steel sheet comprising, by area percentage, 10% to 20% Bainite, 80% to 92% Ferrite, wherein the aspect ratio of the grains of Bainite and

Ferrite is less than 1.75 and a cumulated amount of Residual Austenite and Martensite from 0% to 10%.

17. The hot rolled steel sheet according to claim **16** wherein the composition includes 0.35% to 0.75% of Silicon.

18. The hot rolled steel sheet according to claim **16** wherein the composition includes 0.065% to 0.1% of Carbon.

19. The hot rolled steel sheet according to claim **16** wherein the composition includes 0.02% to 0.07% of Niobium.

20. The hot rolled steel sheet according to claim **16** wherein the composition includes 1.45% to 1.7% of Manganese.

21. The hot rolled steel sheet according to claim **16** wherein the composition includes 0.01% to 0.06% of Titanium.

22. The hot rolled steel sheet according to claim **16** wherein the composition includes 0.02% to 0.06% of Aluminum.

23. The hot rolled steel sheet according to claim **16** wherein the cumulated amount of Residual Austenite and Martensite is from 0% to 5%.

24. The hot rolled steel sheet according to claim **16** wherein the steel sheet has a hole expansion ratio of 75% or more.

25. A method of production of a hot rolled steel sheet comprising the following successive steps:
 providing a semi-finished product with a composition, expressed in percentage by weight:

0.060% ≤ Carbon ≤ 0.1%;
 1.4% ≤ Manganese ≤ 1.8%;
 0.3% ≤ Silicon ≤ 0.8%;
 0.01% ≤ Aluminum ≤ 0.1%;
 0.01% ≤ Niobium ≤ 0.09%;
 0.01% ≤ Titanium ≤ 0.09%;
 0.002% ≤ Phosphorus ≤ 0.02%;
 0% ≤ Sulfur ≤ 0.005%;
 0% ≤ Nitrogen ≤ 0.01%;

-continued
with

$0.045\% \leq \text{Ti} + \text{Nb} \leq 0.060\%$;

and one or more of the following optional elements:

$0\% \leq \text{Chromium} \leq 0.5\%$;

$0\% \leq \text{Molybdenum} \leq 0.4\%$;

$0\% \leq \text{Vanadium} \leq 0.4\%$;

$0\% \leq \text{Calcium} \leq 0.005\%$;

$0\% \leq \text{Copper} \leq 1\%$;

$0\% \leq \text{Nickel} \leq 1\%$;

$0\% \leq \text{Boron} \leq 0.05\%$;

$0\% \leq \text{Magnesium} \leq 0.05\%$;

a remainder of the composition being composed of iron
and unavoidable impurities caused by processing;
reheating the semi-finished product to a temperature from
1200° C. to 1300° C.;

rolling the semi-finished product in the austenitic range
wherein the hot rolling finishing temperature is from
850° C. to 975° C. to obtain a hot rolled steel strip;
cooling the hot rolled strip is cooled wherein the cooling
starts immediately after the finishing of hot rolling;
then cooling the hot rolled strip from finishing of hot
rolling to a cooling stop temperature range from 480°
C. to 550° C. at a cooling rate greater than 20° C./s;
coiling the hot rolled strip from the coiling temperature
range 480° C. to 550° C.; and
cooling the coiled hot rolled strip to room temperature to
obtain a hot rolled steel sheet.

26. The method according to claim **25** wherein the reheat-
ing temperature for semi-finished product is from 1200° C.
to 1275° C.

27. The method according to claim **25** wherein the hot
rolling finishing temperature is from 880° C. to 930° C.

28. The method according to claim **25** wherein the cooling
rate for the step one of cooling is from 40° C./s to 120° C./s.

29. A method for manufacturing a structural or safety part
of a vehicle comprising employing the steel sheet as recited
in claim **15**.

30. A method for manufacturing a structural or safety part
of a vehicle comprising employing the steel sheet according
to the method as recited in claim **25**.

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